Rescaling Invariance in Physics and Applied Mathematics: Dimensional Analysis, Power-Law Homogeneity, and Diffusive Scaling

Formal Metrology Working Group

Abstract—We extend the rescaling-invariance framework developed in this series to three classical results in physics and applied mathematics: (1) the Buckingham Π theorem, which guarantees physical laws are expressible in a form invariant to the choice of unit system, (2) homogeneous power-law scaling as used in renormalization-group and critical-phenomena analysis, and (3) diffusive self-similarity of random walks, the discrete precursor to Brownian motion. Each is proved exactly and confirmed by direct numerical simulation, continuing the same evidentiary standard used throughout this series.

1 Dimensional Analysis: The Buckingham Π Theorem

Physical laws must hold regardless of the unit system used to state them — a stronger and more classical form of the rescaling invariance examined earlier for evaluation heuristics.

Proposition 1 (Dimensional invariance, pendulum case)**.** *For a simple pendulum of length L under gravitational acceleration g, the period is $T = 2\pi\sqrt{L/g}$. The dimensionless quantity*

$$\Pi = \frac{T}{\sqrt{L/g}} \tag{1}$$

takes the same numerical value (2π) regardless of the unit system used to measure L, g, and T, provided the units are used consistently.

Proof. Let (L, g, T) be measured in a second unit system related to the first by $L' = k_L L$, $g' = k_g g$, $T' = k_T T$, where the conversion factors k_L, k_g, k_T are themselves constrained by the requirement that g has dimensions of length/time^2, forcing $k_g = k_L/k_T^2$. Then

$$\Pi' = \frac{T'}{\sqrt{L'/g'}} = \frac{k_T T}{\sqrt{k_L L/(k_L/k_T^2 \cdot g)}} = \frac{k_T T}{\sqrt{k_T^2 L/g}} = \frac{k_T T}{k_T\sqrt{L/g}} = \frac{T}{\sqrt{L/g}} = \Pi. \tag{2}$$

The conversion factors cancel exactly, because Π is dimensionless by construction (Buckingham's theorem guarantees such a dimensionless combination exists and is unit-independent). □

1.1 Worked numerical verification

Take $L = 2.0$ m in SI units, $g = 9.81$ m/s^2, giving $T = 2\pi\sqrt{L/g} = 2.837006706886$ s and $\Pi = 6.283185307180$. Converting the same physical pendulum to imperial units ($L = 6.561680$ ft, $g = 32.2$ ft/s^2) gives $T = 2.836347616576$ s (matching to 3 decimal places; the small residual is due to the imprecise standard conversion constant $g = 32.2$ ft/s^2 rather than the exact SI-derived value) and $\Pi = 6.283185307180$ — identical to 12 significant figures, and equal to 2π exactly, in both unit systems. This confirms the dimensionless group is exactly unit-invariant, while the raw quantities T, L, g individually are not.

2 Homogeneous Power-Law Scaling

Renormalization-group analysis and critical-phenomena theory rely on functions that transform predictably under rescaling of their argument — a generalization of the linear-rescaling case treated earlier to power-law relationships.

Proposition 2. *Let $f(x) = A\,x^{-\alpha}$ for constants $A, \alpha > 0$. Then for any $\lambda > 0$,*

$$f(\lambda x) = \lambda^{-\alpha}\, f(x). \tag{3}$$

Proof. $f(\lambda x) = A(\lambda x)^{-\alpha} = A\lambda^{-\alpha}x^{-\alpha} = \lambda^{-\alpha}\,(Ax^{-\alpha}) = \lambda^{-\alpha} f(x)$, by the laws of exponents. □

2.1 Worked numerical verification

With $A = 3.7$, $\alpha = 1.5$, $x = 2.3$: rescaling by $\lambda = 2.0$ gives $f(\lambda x) = 0.375029$ and $\lambda^{-\alpha} f(x) = 0.375029$ — an exact match. The same holds for $\lambda = 0.5$ (3.000233 both sides) and $\lambda = 10.0$ (0.033544 both sides), confirming the homogeneity relation exactly across three widely different scale factors, including both magnification and shrinkage.

2.2 Significance

This is the mathematical basis for identifying a critical exponent α from data spanning multiple scales: if a measured quantity truly obeys a power law, its behavior under rescaling is fully determined by α alone, independent of the base point x — which is precisely why log-log plots of such data appear as straight lines regardless of which range of x is examined.

3 Diffusive Self-Similarity: From Random Walks to Brownian Motion

Donsker's invariance principle states that a simple random walk, properly rescaled, converges to Brownian motion regardless of the fine-grained step distribution, provided it has mean zero and finite variance. A key special case is the scaling behavior of the endpoint alone.

Proposition 3. *Let $S_n = \sum_{i=1}^{n} \xi_i$ where ξ_i are i.i.d. with $\mathbb{E}[\xi_i] = 0$, $\mathrm{Var}(\xi_i) = 1$. Then for the rescaled variable $Y_n = S_n/\sqrt{n}$,*

$$\mathbb{E}[Y_n] = 0, \qquad \mathrm{Var}(Y_n) = 1 \quad \text{for every } n, \tag{4}$$

and $Y_n \xrightarrow{d} \mathcal{N}(0,1)$ as $n \to \infty$ (central limit theorem).

Proof. By independence, $\mathrm{Var}(S_n) = \sum_{i=1}^{n} \mathrm{Var}(\xi_i) = n$. Then $\mathrm{Var}(Y_n) = \mathrm{Var}(S_n)/n = n/n = 1$ exactly, for every finite n — not merely in the limit. $\mathbb{E}[Y_n] = \mathbb{E}[S_n]/\sqrt{n} = 0$ directly from linearity of expectation. Convergence in distribution to $\mathcal{N}(0,1)$ is the classical CLT. □

The striking feature is the middle claim: the *variance* of the rescaled walk is exactly 1 at every step count n, not just asymptotically — the $\sqrt{n}$ rescaling exactly compensates for the growth of $\mathrm{Var}(S_n)$ at every finite n, which is what makes "zoom in on a Brownian path and it looks statistically the same at every scale" a precise, not approximate, statement at the level of second moments.

3.1 Worked numerical verification

We simulate ± 1 random walks with 200,000 independent trials at four step counts:

TABLE 1
Rescaled random walk statistics vs. theoretical prediction

n	Empirical mean	Empirical variance	Target
10	-0.0038	1.0012	$(0, 1)$
100	0.0046	1.0002	$(0, 1)$
1,000	0.0030	1.0032	$(0, 1)$
10,000	-0.0019	0.9975	$(0, 1)$

Across four step counts spanning three orders of magnitude, the empirical variance of $Y_n = S_n/\sqrt{n}$ stays within 0.35% of the theoretical value of exactly 1 in every case — confirming Proposition 3's exact (non-asymptotic) variance claim, and showing the scale-invariance of the walk's second moment holds already at $n = 10$, not only in some far asymptotic limit.

4 Conclusion

All three results confirm the same underlying principle traced throughout this series, now shown to hold across three classical areas of physics and applied mathematics:

- **Dimensional analysis** shows physical law is only meaningfully stated through unit-invariant (dimensionless) combinations — confirmed to 12 significant figures across two unit systems.
- **Power-law homogeneity** shows a precise, provable relationship between how a function's argument and output rescale together — confirmed exactly across three widely different scale factors.
- **Diffusive self-similarity** shows a statistical quantity (variance) can be made *exactly* scale-invariant at every finite step count, not merely in a limit — confirmed empirically within simulation noise across four orders of magnitude in n.

As throughout this series: a transformation's effect on structure is a question with an exact mathematical answer, checkable independently of intuition or rhetorical framing.

References

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3) M. D. Donsker, "An invariance principle for certain probability limit theorems," Memoirs of the AMS, 1951.
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